



Module: Electronics, system components

Series of TD n° 01

Exercise 1:

Give the correct answer (A, B, C or D):

1. What is electric current?

- a) An electromagnetic force
- b) An electric field
- c) A movement of electrons in a conducting medium
- d) Electrons that orbit the nuclei of atoms

2. What is an electrical conductor?

- a) A metal
- b) A material which offers only very low resistance to the passage of current
- c) An electronic component

3. What is an electrical insulator?

- a) A non-metal
- b) A material that does not conduct heat
- c) A material which offers very high resistance to the passage of current

4. What phenomena are observed when a current passes through a resistance?

- a) A loss of current between the input and output of the resistor
- b) A light
- c) Heating of the resistance

5. When two resistors are put in parallel?

- a) They are subjected to the same tension
- b) The current is divided into two equal parts
- c) The heat produced is less

6. When two resistors are put in series?

- a) The current in the second is less than in the first
- b) The current is identical in each of them
- c) The current is less in the greater resistance
- d) The voltage is divided into two equal parts

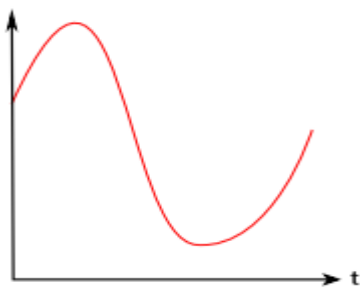
7. Choose propositions that are accurate

- a) Resistance is expressed in Ohm
- b) The volt is the unit of voltage
- c) The ampere is a unit of power

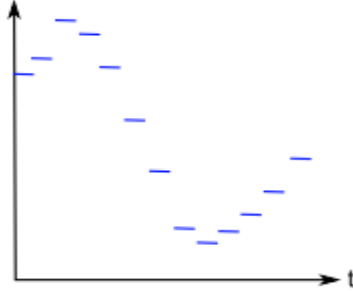
Exercise 2:

- 1. What is the difference between low current and high current? Name an area of use for each of the currents?
- 2. What is the purpose of modulating a signal?

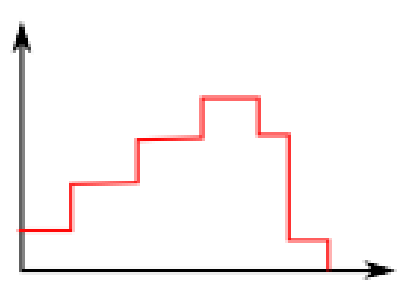
3. What is the difference between analog signals and digital signals?
4. Classify the signals described or represented below as “analog” or “digital”



Change in temperature over the course of a day



Hourly temperature display



Film recorded by a VCR on VHS tape

Film streamed online

Exercise 3:

Give the correct answer (a, b, c or d) for each of the following questions:

1. Frequency is expressed in international units as:

- a) second b) Hertz c) minute d) cycle per second

2. The period is expressed in international units as:

- a) second b) Hertz c) minute d) hour

3. The period is:

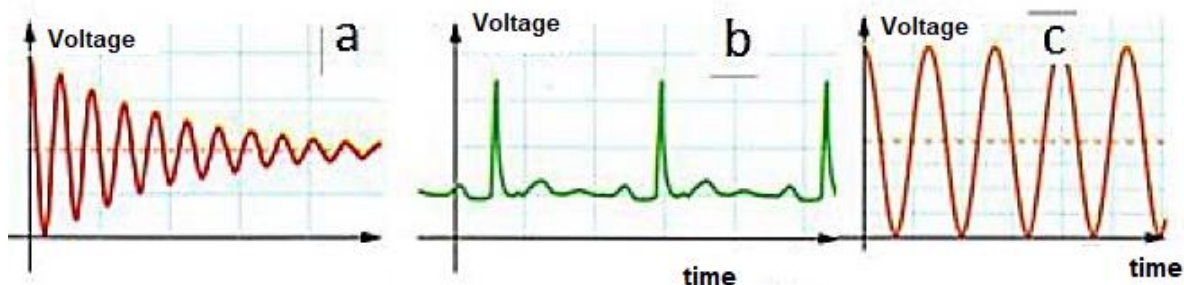
- a) the opposite of frequency b) the same as the frequency c) the inverse of the frequency

4. An event A is more frequent than an event B. So:

- a) The period of A is longer than that of B
 b) The period of B is longer than that of A
 c) A and B have the same period
 d) We cannot know which is longer among the period of A or B.

Exercise 4:

1. What is the definition of period T ? Of the frequency F ?
2. A phenomenon occurs twice per second. What is its frequency?
3. Another phenomenon can be observed every 3 s. Determine its period.
4. The heartbeat rate is 1.33 beats/s. Calculate the period of the beats.
5. Among the signals shown opposite, which are periodic? Justify.

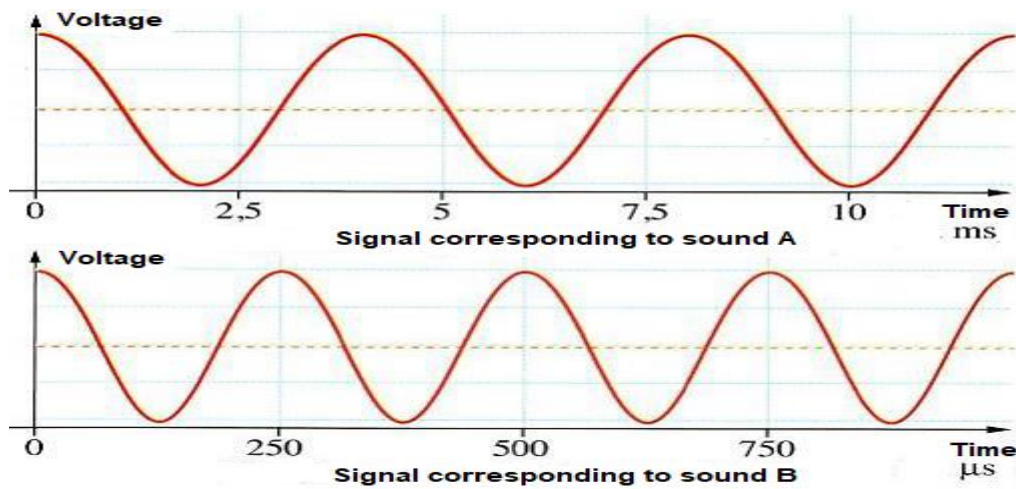


Exercise 5:

During an audiogram, the doctor tests his patient's hearing. The electrical signals corresponding to two sounds A and B emitted during this test are shown in the figure below. Sound A is not perceived by the patient while sound B is well perceived.

1. Determine the periods T_A and T_B of sounds A and B then convert the values into the legal unit of time.

2. Calculate the corresponding FA and FB frequencies.
3. Does the patient have low-frequency or high-frequency hearing problems? Justify.



Exercise 6:

The speed of a modem (old generation) is 56,000 bits/s. A character is encoded on one byte (8 bits).

1. How many characters will be transferred per minute?
2. Now we're talking about broadband. What is the speed of the modem which can load a 1.5 GB* video file in 25 min.

* Go = gigaoctet = 10^9 octets

Exercise 7:

- 1) What is the difference between active electronic components and passive electronic components?
- 2) Report the correct answer (A: for an active component, or B: for a passive component) in the "Answer" column for each of the following basic electronic components.

Components	Answer
1. Vacuum Tubes,	
2. Transistors,	
3. Diodes,	
4. Integrated Circuit (IC),	
5. LED light-emitting diode (Light-Emitting Diode)	
6. Resistance,	
7. Capacitor,	
8. Transformer,	

Good luck
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